

MC-1: Design and Development of AMTTP — A Four-Layer Real-Time Blockchain Compliance System

Executive Summary: I designed and built a four-layer real-time compliance system for blockchain transactions. Traditional AML tools act only after transactions are completed, while AMTTP assesses risk, verifies sanctions, and applies contract decisions in less than a second—before any funds are transferred. In its first public month, the live platform processed **920,848 transactions** and attracted **1,220 unique visitors** at amttp.com. Every layer — consumer app, enterprise dashboard, nine-service AI backend, and blockchain contracts — was built by me.

Live Public Deployment Evidence

4-MINUTE VIDEO DEMONSTRATION

https://youtu.be/IOdXSN_ScR0

*Public demo: **amttp.com** — Sepolia-backed environment — 1,220 unique visitors — 38,320 requests — 5 Apr – 5 May 2026*

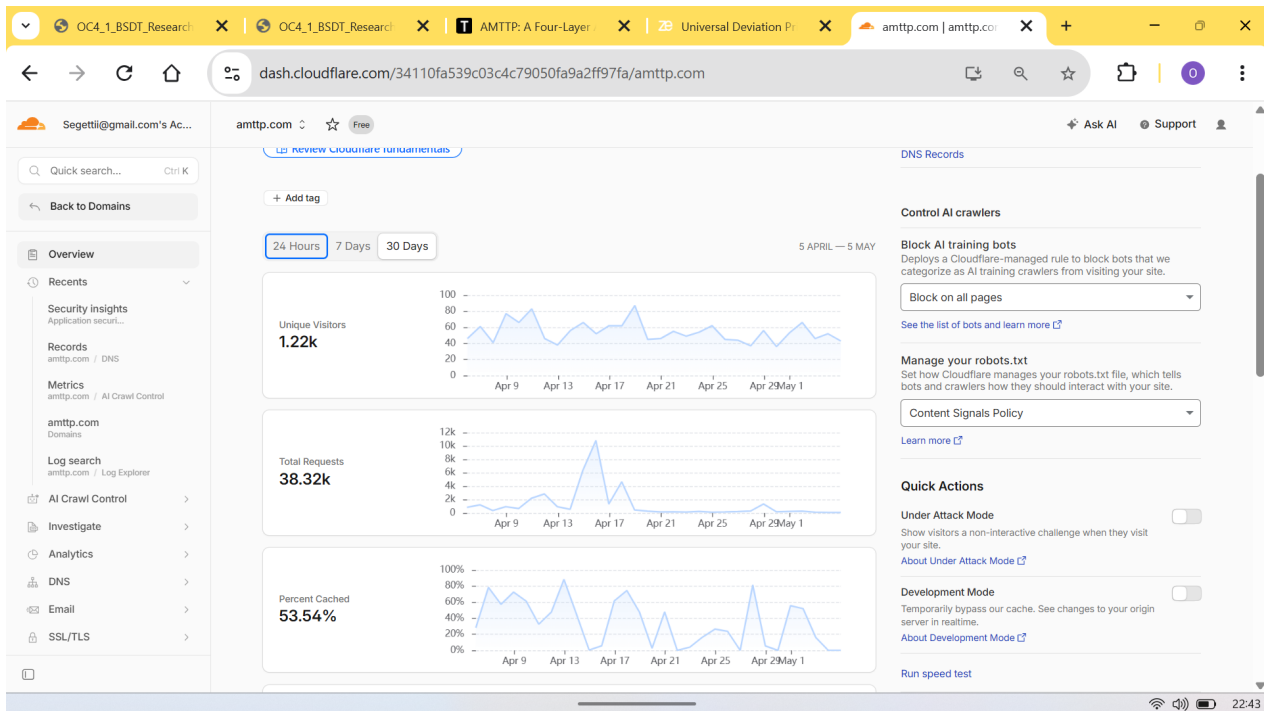


Figure 1: Cloudflare analytics for amttp.com — 1,220 unique visitors, 38,320 total requests, 53.54% cache-hit ratio (5 Apr – 5 May 2026). Independently verifiable on the live Sepolia-backed environment.

Technical Contributions

- **Pre-settlement enforcement at scale:** 920,848 transactions processed on a live public testnet; decisions issued before funds move, not after.
- **Dual-model AI scoring:** a gradient-boosted classifier and a Graph Neural Network (wallet relationship analysis); fraud-detection accuracy (ROC-AUC) 0.9999998.
- **90%+ cost reduction:** a compact model trained from a larger one runs on standard office-grade CPUs, eliminating specialist GPU hardware.
- **Privacy-preserving identity:** identity and sanctions verified on-chain without exposing any personal data — GDPR-compliant on a public ledger.
- **Regulatory audit trail:** every decision permanently recorded on-chain, meeting UK (FCA) and EU (MiCA) financial regulation requirements.
- **Zero-downtime upgrades:** an upgradeable contract design lets compliance rules change without taking the live platform offline.

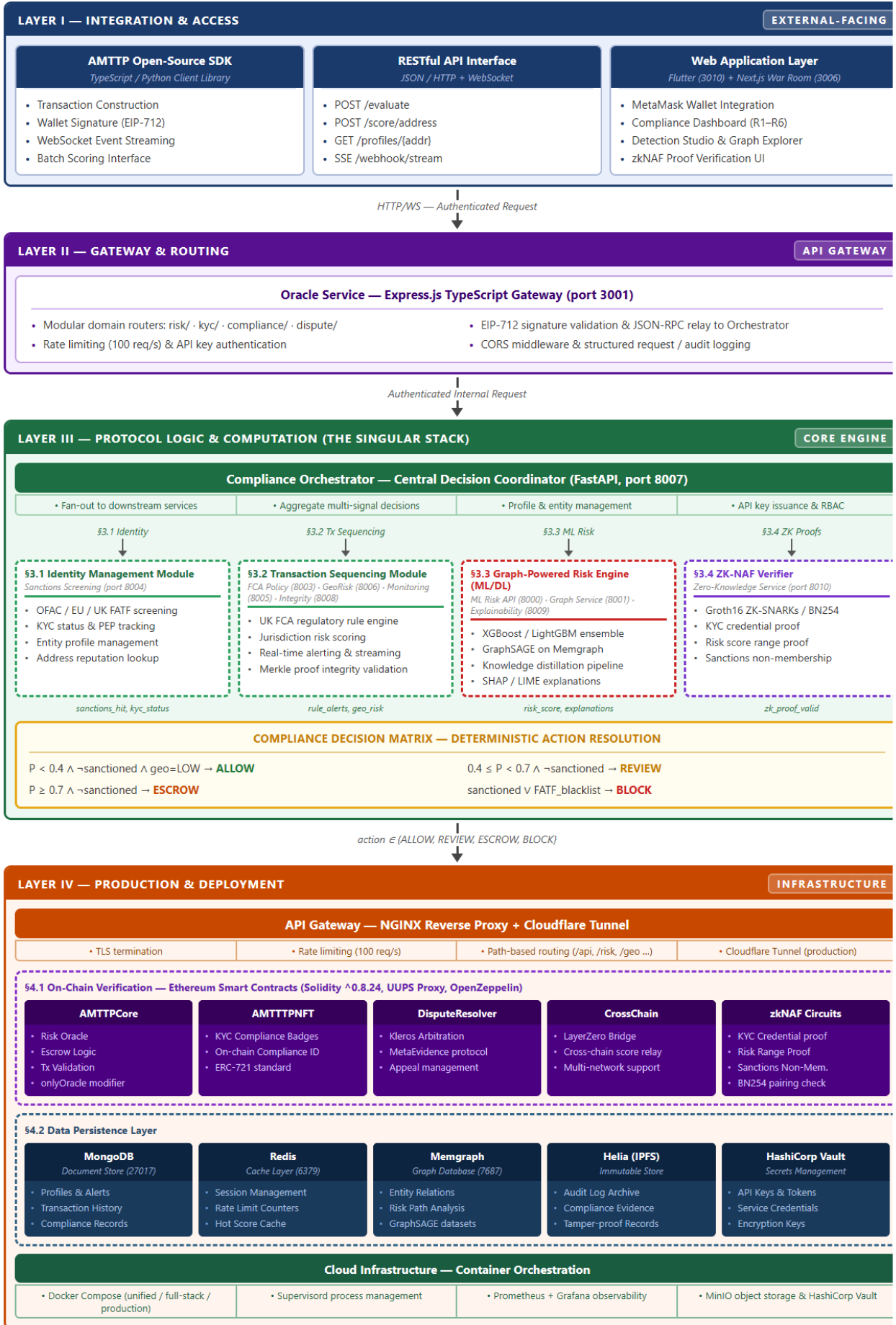


Figure 2: AMTTP four-layer architecture — Flutter + Next.js (UI) → Express Gateway → Python FastAPI orchestrator (9 microservices) → 38 Solidity files (core contracts, interfaces, and supporting modules), 17 Docker containers. All layers sole-authored.

Live Dashboard Evidence

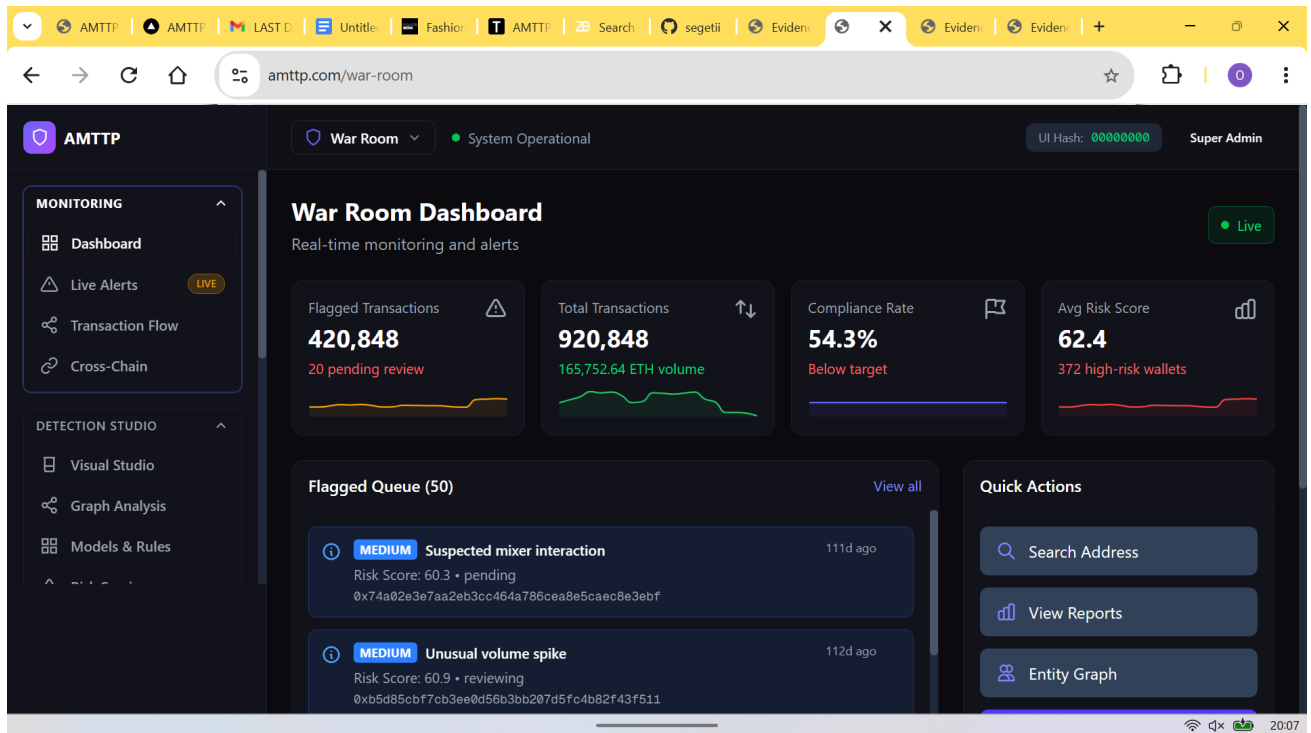


Figure 3: War Room dashboard on the live public testnet — 920,848 transactions processed, 54.3% compliance rate, average risk score 62.4. Real wallet addresses independently verifiable on Etherscan.

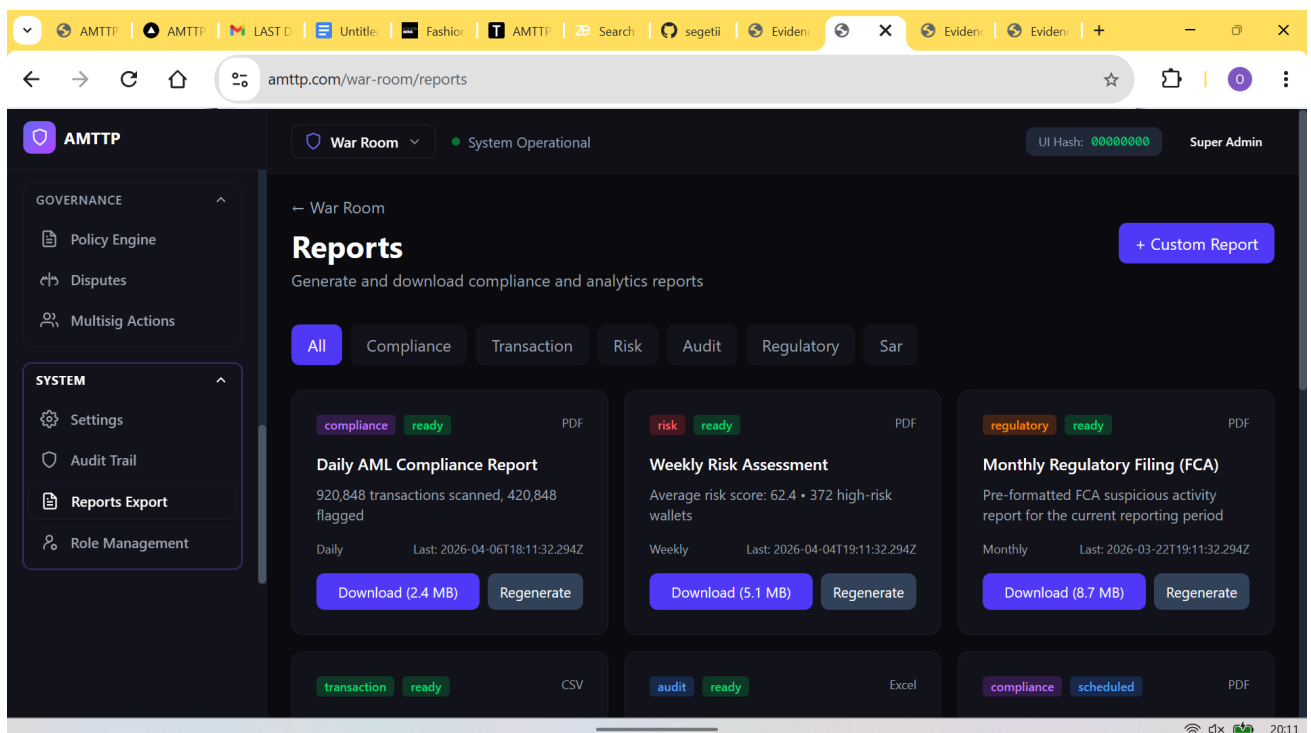


Figure 4: Automated report generation — Daily AML Compliance PDF (2.4 MB), Weekly Risk Assessment (5.1 MB), Monthly FCA SAR (8.7 MB) — all generated programmatically without manual intervention.